B1 AdaLoRA summarizer: Methodology and Evidence

CS781 Capstone – Krishna K Karthikeya G N

12 November 2026

Status. Post-run methodology reconstruction grounded in archived source and reports. This document is not an executed historical notebook.

Research question

Try allocating low-rank adaptation capacity where the summarizer's B1 projections most need it. The dynamic rank budget is a parameter-efficiency experiment; it does not change the task's 100 test IDs or the summary output schema.

Architecture and mechanism

AdaLoRA wraps linear layers with trainable low-rank factors and adjusts rank allocation during optimizer updates. The B1 passage-fusion path supplies long-document context.

Measured result

Scored submission	1030985
Codabench BERTScore F1	0.7656
Archived local metric	-1.6305 validation negative NLL; BERTScore not measured in this report

Codabench BERTScore F1 was 0.7656. The archived report selected epoch 1, so this result is particularly sensitive to training duration and checkpoint selection.

Limitations and next check

The saved evidence does not support a claim that adaptive rank allocation beat the fixed B1 fusion system. Local, pilot, and hidden metrics are deliberately kept distinct. A single scored run does not establish a causal architecture effect or statistical significance.

Source and reproducibility

Original source modules: b1_adalora.py, b1_fusion.py, train_phase6.py. SHA-256 hashes, the original report, and the prediction ZIP receipt are in the companion provenance file. The explanatory notebook was written after the historical run; original execution output is never implied. Reproduction requires authorized course data, the Phase 1 parent weights/tokenizer, and the dependencies in the archived source.

Documentation disclosure

AI assistance was used to organize the archived evidence and draft this explanatory document. All numerical claims above come from saved run records or the provided finished-submission screenshot; missing evidence is labelled as such.